



UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Advanced Level

MATHEMATICS

9709/32

Paper 3 Pure Mathematics 3 (P3)

May/June 2011

1 hour 45 minutes

Additional Materials: Answer Booklet/Paper
Graph Paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 75.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

This document consists of **3** printed pages and **1** blank page.



- 1 Solve the inequality $|x| < |5 + 2x|$. [3]

- 2 (i) Show that the equation

$$\log_2(x + 5) = 5 - \log_2 x$$

can be written as a quadratic equation in x . [3]

- (ii) Hence solve the equation

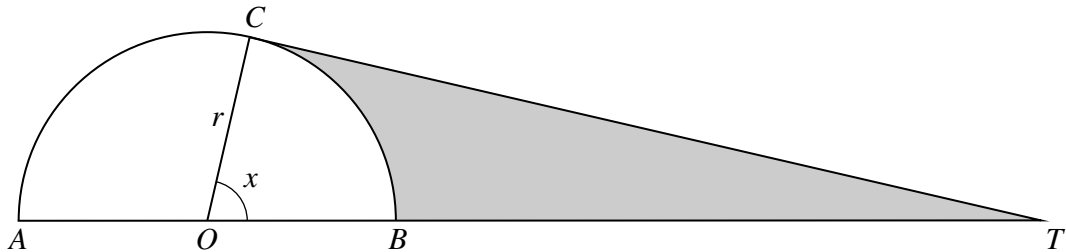
$$\log_2(x + 5) = 5 - \log_2 x. \quad [2]$$

- 3 Solve the equation

$$\cos \theta + 4 \cos 2\theta = 3,$$

giving all solutions in the interval $0^\circ \leq \theta \leq 180^\circ$. [5]

4



The diagram shows a semicircle ACB with centre O and radius r . The tangent at C meets AB produced at T . The angle BOC is x radians. The area of the shaded region is equal to the area of the semicircle.

- (i) Show that x satisfies the equation

$$\tan x = x + \pi. \quad [3]$$

- (ii) Use the iterative formula $x_{n+1} = \tan^{-1}(x_n + \pi)$ to determine x correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

- 5 The parametric equations of a curve are

$$x = \ln(\tan t), \quad y = \sin^2 t,$$

where $0 < t < \frac{1}{2}\pi$.

- (i) Express $\frac{dy}{dx}$ in terms of t . [4]

- (ii) Find the equation of the tangent to the curve at the point where $x = 0$. [3]

- 6 A certain curve is such that its gradient at a point (x, y) is proportional to xy . At the point $(1, 2)$ the gradient is 4.

(i) By setting up and solving a differential equation, show that the equation of the curve is $y = 2e^{x^2-1}$. [7]

(ii) State the gradient of the curve at the point $(-1, 2)$ and sketch the curve. [2]

- 7 (a) The complex number u is defined by $u = \frac{5}{a+2i}$, where the constant a is real.

(i) Express u in the form $x + iy$, where x and y are real. [2]

(ii) Find the value of a for which $\arg(u^*) = \frac{3}{4}\pi$, where u^* denotes the complex conjugate of u . [3]

- (b) On a sketch of an Argand diagram, shade the region whose points represent complex numbers z which satisfy both the inequalities $|z| < 2$ and $|z| < |z - 2 - 2i|$. [4]

- 8 (i) Express $\frac{5x - x^2}{(1+x)(2+x^2)}$ in partial fractions. [5]

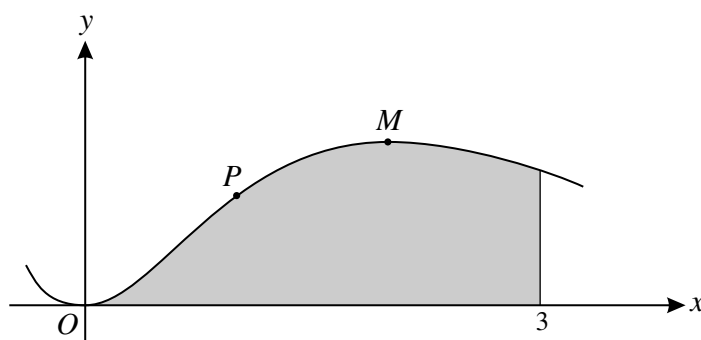
(ii) Hence obtain the expansion of $\frac{5x - x^2}{(1+x)(2+x^2)}$ in ascending powers of x , up to and including the term in x^3 . [5]

- 9 Two planes have equations $x + 2y - 2z = 7$ and $2x + y + 3z = 5$.

(i) Calculate the acute angle between the planes. [4]

(ii) Find a vector equation for the line of intersection of the planes. [6]

10



The diagram shows the curve $y = x^2 e^{-x}$.

(i) Show that the area of the shaded region bounded by the curve, the x -axis and the line $x = 3$ is equal to $2 - \frac{17}{e^3}$. [5]

(ii) Find the x -coordinate of the maximum point M on the curve. [4]

(iii) Find the x -coordinate of the point P at which the tangent to the curve passes through the origin. [2]

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.